

# Stage 2 Behavioral Feature Design — C22 veNTUre

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Grounded in `notebooks/profiling.ipynb` (46,520 trades · 502 accounts · campaigns 33–66) and the business context in `C22_Explained.pdf`. Every feature below is checked against what the dataset can *actually* support — no starting balance, no pass/fail label, no entry-fee/pricing data, single instrument (XAUUSD), and no tick/path data between open and close. Features that would require any of those are called out as infeasible rather than invented.

### 0. Grounding: how C22 makes money, and what Stage 1 already closed off

**Revenue:** commission (−\$299,240, visible, mechanical, `amount` ↔ `commission`  $r = -1.00$ ) + mirror-book P&L (+\$11,834, `reverseProfit`  $\approx -$  `profit`) + entry fees (not in the data). Commission is ~25× the directional result — **C22’s business is driven by trade frequency and survival, not by beating traders directionally**. That reframes what “behavioral feature” should mean here: not “does this trait predict who trades gold well” but “does this trait predict who keeps paying commission, and who blows up before they can.”

**Key business metrics this dataset can actually speak to:**

| Metric          | Proxy available                                                                         |
|-----------------|-----------------------------------------------------------------------------------------|
| Profitability   | <code>sign(sum(netProfit))</code> per account (explicitly a proxy — no pass/fail label) |
| Retention / LTV | campaigns-participated per account, session survival time                               |
| Risk exposure   | max drawdown from own peak, no-SL rate                                                  |

| Metric            | Proxy available                                       |
|-------------------|-------------------------------------------------------|
| Churn             | single-campaign accounts vs. multi-campaign accounts  |
| Campaign response | behavior deltas across campaigns for the same account |

**What Stage 1 already tested and killed — do not resubmit these as raw aggregates:** trade count, average lot size, median duration, no-SL rate, total commission (all  $p > 0.2$  against outcome), round-number price anchoring (not supported), disposition effect by raw duration ( $p = 0.56$ ), loss-chasing by size (significant but backwards, ~1.6% effect). The one survivor was **drawdown depth from own peak** ( $p < 0.001$ ), and the one secondary lead was **re-entry haste after a loss**. Every feature family below is built to either (a) decompose *why* drawdown separates outcomes, (b) extend the haste lead with the rigor Stage 1 flagged as missing, or (c) surface a genuinely new latent trait Stage 1's population-level cuts couldn't see.

**Explicitly infeasible with this dataset (do not build, and reject if proposed elsewhere):**

- Portfolio concentration / diversification discipline — single instrument (XAUUSD only), no asset mix exists to be concentrated or diversified.
- Profit give-back / MAE-MFE / “distance from intraday high before close” — no tick or path data, only open/close snapshots.
- True herding on *direction* — no evidence traders can observe each other's positions; only trade-*timing* coincidence is measurable, and it's a weak proxy at best.
- Anything requiring starting balance, entry fee, or pass/fail label (LTV in dollars, true drawdown-as-%-of-target, true churn vs. pass).

## 1. Feature Catalog (30 features, organized by family)

Each feature is a **per-account, cross-trade or cross-campaign trait** — not a single-session or single-campaign number — because a trait has to be stable enough to act on. “Stability” below means: computed from  $\geq 10$  trades or  $\geq 2$  campaigns per account (the same threshold Stage 1 used for its own no-SL-rate cut), and expected to persist because it reflects a habitual response pattern, not a one-off market outcome.

### Family: Risk Behavior

#### 1. Stop-Loss Elasticity (Risk Shape Score)

- *Theory:* Prospect Theory (fat-tail risk-seeking)
- *Definition:*  $(\text{median}(\text{netProfit} \mid \text{no-SL}) - \text{median}(\text{netProfit} \mid \text{has-SL})) / \text{abs}(\min(\text{netProfit} \mid \text{no-SL}))$ , computed per account with  $\geq 10$  no-SL trades

- *Captures*: whether *this* account's no-SL habit reproduces the population's fat-tail signature (small typical edge, rare catastrophe) in a proportionate way, or whether their tail loss dwarfs their typical edge
- *Stability*: a structural habit (how far a trader lets a losing position run), not a market-condition artifact
- *Business insight*: separates "small edge, contained tail" traders (fine to leave alone) from "small edge, unbounded tail" traders (exactly who a catastrophe-only stop protects)
- *Intervention*: mandatory wide catastrophe-only stop for accounts scoring above a tail threshold
- *Priority*: **High Value**

## 2. Leverage Utilization Consistency

- *Theory*: Trading Discipline / Capital Allocation
- *Definition*:  $\text{std}(\text{amount}) / \text{mean}(\text{amount})$  per account (coefficient of variation of lot size)
- *Captures*: rules-based, repeatable sizing vs. erratic/situational sizing
- *Stability*: a sizing habit, shown in Stage 1 to hold even for the busiest 15 accounts (5× spread in medians, wide within-account spread too) — this feature formalizes that observation into a comparable score
- *Business insight*: low-CV accounts are predictable revenue (steady commission); high-CV accounts are volatility risk on both sides (mirror-book exposure and drawdown)
- *Intervention*: none direct — a segmentation input, not a nudge target on its own
- *Priority*: **Nice to Have**

## 3. Risk Escalation Slope

- *Theory*: Risk Seeking in Losses / Decision Fatigue
- *Definition*: per-session regression slope of  $\text{amount}$  on hours-into-session, averaged across an account's sessions
- *Captures*: whether a trader's *own* sizing trajectory rises (escalating risk as the session wears on) or falls (fatigue/caution) — Stage 1 found the *population* slope is negative ( $0.196 \rightarrow 0.12$ ); this turns it into a per-account score so outliers (traders who escalate against the population trend) become visible
- *Stability*: averaging across sessions turns a single-session pattern into a trait
- *Business insight*: identifies the minority who escalate late-session, the highest-risk segment for a catastrophic single trade
- *Intervention*: session-length nudge or forced cool-down for late-session, high-slope accounts
- *Priority*: **High Value**

## 4. Catastrophic Exposure Rate

- *Theory*: Fat-Tail Risk

- *Definition:* % of an account's own no-SL trades whose loss exceeds 3× that account's own median loss
- *Captures:* personalized tail exposure — normalized to the trader's own scale rather than the population's −\$944 outlier, so it's comparable across traders with very different typical sizes
- *Stability:* a recurring exposure pattern, not one bad trade (requires ≥2 qualifying trades to score)
- *Business insight:* a cleaner intervention trigger than a fixed-dollar threshold, since it adapts to each account's own size
- *Intervention:* per-account catastrophe stop calibrated to their own historical tail
- *Priority:* **High Value**

## Family: Emotional Trading / Behavior Under Pressure

### 5. Loss Reactivity Ratio (Recovery Chasing Score)

- *Theory:* Prospect Theory / Loss Aversion
- *Definition:* `median(amount | prev_outcome = after_loss) / median(amount | prev_outcome = after_win)` , per account (≥5 of each, matching Stage 1's own eligibility bar in §5.5)
- *Captures:* Stage 1 already ran this test **at the population level** (0.179 vs 0.182,  $p = 0.0058$ , ~1.6% — rejected as economically negligible). This is the same construct computed **per account** — turning one population statistic into a scorable trait, which is the natural next step, not a duplicate: a population median near 1.0 is fully compatible with a wide spread of individual ratios, some genuinely loss-chasing, some genuinely loss-averse, canceling out in aggregate.
- *Stability:* a within-account paired comparison, same logic Stage 1 validated as appropriate for this data (non-independent trades)
- *Business insight:* if the *individual* distribution is wide even though the *population* average is flat, C22 can still target the tail of genuine chasers instead of concluding “no one chases losses”
- *Intervention:* targeted cool-off prompt only for accounts scoring materially above 1.0, not a blanket rule
- *Priority:* **Must Have**

### 6. Re-entry Haste Index

- *Theory:* Regret Aversion / Attention Bias
- *Definition:* `median(reentry_gap_min | after_loss) / median(reentry_gap_min | after_win)` , per account
- *Captures:* the account-level version of Stage 1's H4 (population: 3.5 min vs 5.1 min) — this is *literally* the feature the hypothesis document calls for under “Stage-2 test” for H4
- *Stability:* re-entry timing habit, stable enough to average across an account's full trade history

- *Business insight:* identifies which accounts' pressure response is haste-driven, the one channel Stage 1 found real signal in
- *Intervention:* mandatory pause timer after a loss, scaled to how far below 1.0 an account's own index sits
- *Priority:* **Must Have**

## 7. Tilt Score (consecutive-loss escalation)

- *Theory:* Loss Aversion / Recency Bias ("tilt")
- *Definition:* slope of `amount` (or `1/reentry_gap_min`) against consecutive-loss streak length (1, 2, 3, 4+), per account
- *Captures:* what a single-lag comparison (Stage 1's whole approach) structurally cannot: escalating reaction to a *run* of losses, not just the one before. This is a genuinely new construct, not a rerun of the rejected loss-chasing hypothesis — Stage 1 only ever tested lag-1.
- *Stability:* requires accounts with enough multi-loss streaks ( $\geq 3$  streaks of length  $\geq 2$ ) to score reliably; naturally self-filters to active accounts
- *Business insight:* a trader who is flat at lag-1 but escalates sharply after 3+ losses in a row is invisible to Stage 1's test and exactly who a streak-based intervention should target
- *Intervention:* streak-triggered (not just loss-triggered) cool-off
- *Priority:* **High Value**

## 8. Post-Loss Session Abandonment

- *Theory:* Regret Aversion
- *Definition:* P(no further trades in that session | a top-decile loss occurred), per account, averaged across sessions
- *Captures:* withdrawal/regret response to a large loss vs. continuing to trade through it
- *Stability:* requires  $\geq 3$  qualifying large-loss events per account
- *Business insight:* abandoners under-trade relative to their remaining session (revenue left on the table for C22 *and* possibly the safer outcome for the trader); persisters are the drawdown-compounding risk H1 describes
- *Intervention:* a "your session isn't over" re-engagement nudge for abandoners; a pause offer for persisters
- *Priority:* **Nice to Have**

## Family: Discipline

## 9. Stop-Loss Consistency (situational vs. dispositional habit)

- *Theory:* Habit Formation
- *Definition:* `std(no_sl_pct across the campaigns an account participated in)`, per account with  $\geq 3$  campaigns
- *Captures:* Stage 1 found no-SL usage is "a dial, not a switch" population-wide, and explicitly asked whether it's a fixed trait or a per-trade choice (§5.2) — this is the

direct, formal answer at the individual level: low std = a stable personal habit; high std = context-dependent (news events, specific campaigns, mood)

- *Stability*: by construction, this *is* the stability measure — it's the feature that resolves Stage 1's own open question
- *Business insight*: reconciles the account-level null result (50% threshold split,  $p = 0.47$ ) with a still-real underlying effect — it may not be *whether* an account uses stops on average, but *how consistently*, that matters
- *Intervention*: education targeted at high-volatility (situational) accounts specifically, not the whole population
- *Priority*: **Must Have**

## 10. Plan Adherence Score

- *Theory*: Trading Discipline
- *Definition*: % of trades with `slPrice` / `tpPrice` set that actually closed within a small tolerance band of that price, vs. closed materially away from it (discretionary override)
- *Captures*: whether *setting* a stop/target reflects genuine plan-following or is just a cosmetic field a trader ignores in practice
- *Stability*: a distinct construct from “has\_SL rate” — measures honoring a plan, not just having one
- *Business insight*: an account with a high has-SL rate but low adherence isn't actually disciplined — this catches that gap
- *Intervention*: none direct; a segmentation refinement for the drawdown/no-SL story
- *Priority*: **Experimental** (needs confirmation that `closePrice` reliably lands at `slPrice` / `tpPrice` when those triggered — verify against a sample before trusting this at scale)

## 11. Conviction Consistency (side-flip rate)

- *Theory*: Overconfidence / Anchoring
- *Definition*: rate at which consecutive trades on an account switch `side` (BUY↔SELL), relative to that account's own baseline direction split
- *Captures*: sticking with a directional view vs. flip-flopping — a genuinely different construct from the population's balanced 50.5/49.5 BUY/SELL split (which says nothing about any one account's own consistency)
- *Stability*: averaged across an account's full history
- *Business insight*: frequent flippers are the clearest over-trading / low-conviction signal, tying directly into the friction finding (H2) — every flip is another commission event without even a directional thesis behind it
- *Intervention*: fee-awareness nudge specifically framed around flip-driven cost
- *Priority*: **High Value**

## Family: Capital Management / Profit-Taking / Loss Behavior

## 12. Win/Loss Duration Asymmetry (account-level Disposition Ratio)

- *Theory:* Disposition Effect
- *Definition:* `median(durationSec | netProfit > 0) / median(durationSec | netProfit < 0)` , per account
- *Captures:* Stage 1 explicitly flagged this exact follow-up as unfinished: “a within-account version of this test... is a reasonable Stage 2 follow-up before fully closing the door on it.” This is that follow-up, not a duplicate — Stage 1 only ever tested the population median (295s vs 281s,  $p = 0.56$ ), which says nothing about whether individual accounts show the effect in offsetting directions.
- *Stability:* ratio form makes it comparable across accounts with very different absolute holding times
- *Business insight:* if a meaningful subset of accounts show ratio  $\gg 1$  despite the flat population average, that subset is a targeted disposition-effect population worth its own intervention
- *Intervention:* “your losing trades run 3× longer than your winners” personalized feedback
- *Priority:* **High Value**

### 13. Take-Profit Reliance Ratio

- *Theory:* Mental Accounting (asymmetric protection of entry vs. exit)
- *Definition:* `has_TP rate / has_SL rate` , per account
- *Captures:* traders who protect downside but let upside run uncapped (ratio  $\ll 1$ ) vs. those who cap gains but not losses (ratio  $\gg 1$ ) — a genuinely different risk-management style question than either rate alone
- *Stability:* averaged across an account’s trades
- *Business insight:* the  $\gg 1$  group are the disposition-effect and no-SL-tail risks compounding together — a priority segment for the catastrophe-only-stop recommendation
- *Intervention:* pairs naturally with the H3 catastrophe-stop recommendation, targeted at this segment first
- *Priority:* **Nice to Have**

## Family: Time & Routine / Attention & Reactivity

### 14. Session Fatigue Slope

- *Theory:* Decision Fatigue
- *Definition:* per-session regression slope of `no_sl_pct` (or `amount` ) against hours-into-session, averaged across an account’s sessions
- *Captures:* the account-level counterpart of Stage 1’s population finding that lot size shrinks late-session — but applied to risk-management *quality* (no-SL rate) rather than just size, which Stage 1 never tested at the per-account or per-hour level
- *Stability:* averaging across sessions
- *Business insight:* if no-SL rate *rises* late-session for a given account (fatigue-driven shortcutting) even as their size falls, that’s a distinct, previously invisible risk signal

- *Intervention:* a late-session stop-loss reminder for high-slope accounts specifically
- *Priority:* **High Value**

## 15. Cadence Volatility (Attention-Driven Trading Score)

- *Theory:* Attention Bias
- *Definition:* `std(reentry_gap_min) / mean(reentry_gap_min)` , per account
- *Captures:* bursty, attention-triggered trading (long idle stretches, then rapid clusters) vs. steady, metronomic trading — a different construct from the median re-entry gap itself (which only measures typical pace, not consistency of pace)
- *Stability:* averaged across an account's full trade history
- *Business insight:* high-volatility cadence accounts are the ones a “you’ve placed 5 trades in 10 minutes” nudge would actually reach; steady traders wouldn’t trigger it and shouldn’t be bothered by it
- *Intervention:* a rate-limit nudge gated on this score, not applied blanket
- *Priority:* **Nice to Have**

## 16. Early-Session Concentration Ratio

- *Theory:* Attention Bias / Urgency
- *Definition:* % of an account's own trades placed in the first 25% of their own session duration, averaged across sessions
- *Captures:* individual urgency — Stage 1 found ~70% of *all* trades happen in the first 4 hours population-wide; this normalizes to each account's own session length and asks whether *that specific account* front-loads more or less than the population norm
- *Stability:* averaged across sessions
- *Business insight:* extreme front-loaders are the accounts most likely to have already made their decisive (good or bad) trades before any mid-session intervention could reach them — a targeting constraint for the whole early-warning system
- *Intervention:* shifts *when* interventions need to fire for this segment (earlier, not at hour 4)
- *Priority:* **Nice to Have**

## 17. Peak-Hour Habit Consistency

- *Theory:* Habit Formation
- *Definition:* entropy of an account's trade timestamps across hour-of-day, across all campaigns
- *Captures:* whether a trader shows up at consistent times (low entropy — a scheduled habit) or erratically (high entropy)
- *Stability:* requires  $\geq 2$  campaigns to compute meaningfully
- *Business insight:* habitual-hour traders are easier to reach with time-targeted nudges (right message, right hour)
- *Intervention:* schedule nudges/education to an account's own peak hour rather than a fixed broadcast time
- *Priority:* **Experimental**

## Family: Campaign Response / Loyalty & Retention / Learning & Adaptability

### 18. Campaign Recurrence Rate

- *Theory*: Retention (not a classical behavioral-economics theory, but the business's actual LTV proxy)
- *Definition*: campaigns participated in / campaigns available since the account's first appearance
- *Captures*: Stage 1 found repeat participation exists (up to 26 campaigns for some accounts) but only as a raw count; normalizing by *campaigns available since first seen* makes it comparable across accounts that joined at different points, which raw counts cannot do
- *Stability*: by definition a rate, not a one-off count
- *Business insight*: the single cleanest LTV proxy available in this dataset, per the business document's own argument that "survival time is the product" — directly answers which accounts are worth the CAC C22 already spent
- *Intervention*: the composite input for any retention/LTV model
- *Priority*: **Must Have**

### 19. Comeback Latency

- *Theory*: Regret Aversion / re-engagement dynamics
- *Definition*: gap (in campaign IDs) between a losing campaign and an account's next campaign attempt, averaged per account
- *Captures*: do traders come back quickly after a loss (resilience / addiction-adjacent re-engagement) or do they need a long gap (discouragement) — a genuinely different retention signal than raw recurrence rate, since it's about *timing* of return, not *whether*
- *Stability*: requires  $\geq 2$  losing-campaign-then-return events per account
- *Business insight*: short-latency returners after losses are the highest-frequency commission source but also the highest churn-then-burnout risk; useful to know which
- *Intervention*: pacing of "come back" marketing based on this score rather than a uniform cadence
- *Priority*: **Nice to Have**

### 20. Cross-Campaign Behavioral Drift (Learning Adaptability Score)

- *Theory*: Exploration vs. Exploitation / Learning
- *Definition*: slope of (no-SL rate, or max drawdown depth) across an account's successive campaigns, in participation order
- *Captures*: whether repeated attempts show genuine improvement (declining risk exposure — learning) or flat/worsening behavior (no learning, or tilt entrenchment) — this is only possible *because* Stage 1 joined all 34 campaigns, and it's a use of that join Stage 1 itself never made (it validated population-level stability across campaigns, but never asked whether *individual* accounts change over their own repeated attempts)

- *Stability*: requires  $\geq 3$  campaigns per account to fit a slope
- *Business insight*: directly tests whether the challenge format actually teaches discipline over repeated attempts, or just repeatedly extracts fees from unchanging behavior — a genuinely new, structurally-enabled-only-now question
- *Intervention*: if flat/worsening dominates, that's the strongest case for proactive education rather than relying on "learning by doing"
- *Priority*: **High Value**

## 21. Session Survival Time (Endurance Ratio)

- *Theory*: directly operationalizes the business document's "survival time is the product" framing
- *Definition*: (hours from first to last trade in a campaign) / (~28.6h reference window from the blueprint), averaged per account across their campaigns
- *Captures*: how much of the available window an account actually uses before their session effectively ends (by choice or by breach)
- *Stability*: averaged across sessions
- *Business insight*: ties as directly as any feature in this catalog to commission revenue (more surviving hours  $\approx$  more trades  $\approx$  more commission) *and* to the retention story (survivors are more likely to return, per the business document's own argument)
- *Intervention*: the direct target metric for any "keep them at the table" feature — e.g. if a cooling-off nudge extends average survival time without hurting recurrence, it's a win by C22's own logic
- *Priority*: **Must Have**

## 22. Drawdown Recovery Ratio

- *Theory*: directly extends H1
- *Definition*: for accounts whose equity fell  $\geq 50\%$  of their own eventual max drawdown at some point mid-session, the % of that drawdown clawed back before session end
- *Captures*: H1 measures *how deep* an account falls; this measures whether they *come back* — a distinct and more actionable question, since "falls deep but recovers" and "falls deep and stays down" are the same by H1's metric alone but very different for C22's decision of whether to intervene
- *Stability*: requires  $\geq 2$  qualifying drawdown events per account
- *Business insight*: directly informs whether an early-warning intervention should expect a "let it play out" or "must intervene now" response from a given account
- *Intervention*: the account-level input for the H1 threshold-alert design
- *Priority*: **Must Have**

## 23. Warning-Response Simulation Score

- *Theory*: directly operationalizes the H1 recommendation for backtesting
- *Definition*: % of an account's trades placed *after* their equity first crossed a hypothetical drawdown-alert threshold (e.g., 50% of that account's own historical max drawdown) within a session

- *Captures*: literally simulates “what would have happened if the proposed early-warning system had already existed” — this is the feature C22 needs to backtest the H1 recommendation before building it, not a restatement of drawdown depth itself
- *Stability*: requires enough qualifying sessions to trust the rate
- *Business insight*: if this score is materially higher for eventual losers than winners, it validates that the alert would have fired with real lead time, not just at the moment of failure
- *Intervention*: the direct evaluation metric for the proposed drawdown early-warning feature
- *Priority*: **Must Have**

## 24. Exploration vs. Exploitation Score (lot-size entropy)

- *Theory*: Exploration vs. Exploitation
- *Definition*: Shannon entropy of an account’s `amount` values, discretized to the round-increment “menu” Stage 1 identified (0.01, 0.05, 0.1, 0.2, 0.3, 0.4, 0.5, 0.63, etc.)
- *Captures*: Stage 1 found traders “pick from a mental shortlist of round sizes, but which size they pick each time is situational, not fixed” (§5.2) — this turns that qualitative observation into a scorable index: low entropy = habitually exploits one or two sizes; high entropy = explores broadly across the menu
- *Stability*: averaged across an account’s full trade history
- *Business insight*: connects sizing behavior to the friction story (H2) — narrow exploiters may have more predictable, and possibly lower, avoidable commission drag than broad explorers
- *Intervention*: none direct; a segmentation input for friction-reduction messaging
- *Priority*: **Nice to Have**

## 25. Overconfidence Index (Post-Win Escalation)

- *Theory*: Overconfidence
- *Definition*: `median(amount | after 2+ consecutive wins) / baseline median(amount)`, per account
- *Captures*: the mirror image of loss-chasing, and genuinely untested by Stage 1 — Stage 1 tested post-loss sizing exhaustively and rejected it, but never tested whether winning streaks specifically drive size-up, which is the classic overconfidence mechanism, distinct from loss-chasing
- *Stability*: requires  $\geq 5$  qualifying win-streak trades per account
- *Business insight*: if this survives where post-loss sizing didn’t, it reframes the entire “does pressure change sizing” story around *success*, not failure — a materially different, and currently unexplored, angle
- *Intervention*: a “you’ve sized up after wins — stay consistent” nudge, only for high scorers
- *Priority*: **High Value**

## 26. Recency-Weighted Reactivity

- *Theory*: Recency Bias

- *Definition:* correlation of **amount** on trade N with **netProfit** on trade N-1 specifically, vs. correlation of **amount** on trade N with the trailing 5-trade average **netProfit**, per account
- *Captures:* whether sizing overreacts to the single most recent outcome relative to a trader's own recent track record — a sharper and more specific mechanism than the single-lag test Stage 1 already ran, because it directly contrasts recency-weighting against integration
- *Stability:* requires enough trades per account ( $\geq 15$ ) to compute two rolling correlations reliably
- *Business insight:* separates traders reacting to noise (most recent trade) from traders reacting to signal (a real recent slump/streak) — different intervention targets
- *Intervention:* messaging tuned to “don't overreact to one trade” vs. “your last 5 trades suggest a real issue”
- *Priority:* **Experimental**

## 27. Mental Accounting Reset Flag

- *Theory:* Mental Accounting
- *Definition:* for accounts with  $\geq 3$  campaigns, whether opening-session sizing/no-SL rate resets to that account's own long-run baseline each campaign regardless of how the prior campaign ended (vs. carrying over elevated risk from a prior bad campaign into the next one's opening trades)
- *Captures:* whether traders treat each campaign as a separate mental account (consistent with Mental Accounting theory) or as continuous capital/emotional state
- *Stability:* requires  $\geq 3$  campaign transitions per account
- *Business insight:* if resets are the norm, campaign boundaries are a natural, already-existing “clean slate” moment C22 could use for interventions (education, nudges) with minimal resistance
- *Intervention:* time any behavioral messaging to campaign-start, when traders are demonstrably most receptive to a reset
- *Priority:* **Experimental**

## 28. Herding / Crowd-Timing Score (*weak proxy — flagged accordingly*)

- *Theory:* Herding
- *Definition:* correlation between an account's own trade-opening timestamps and the campaign's aggregate trade-volume curve over time
- *Captures:* only *timing coincidence* with the crowd — **not** true herding on direction or price, since nothing in this dataset shows traders observing each other's positions. This must be presented as a weak, indirect proxy, not evidence of copying behavior.
- *Stability:* averaged across campaigns
- *Business insight:* limited — mostly useful as a control variable (to check other features aren't just re-detecting “trades harder when everyone trades harder,” e.g. the front-loaded first-4-hours effect) rather than as a standalone insight
- *Intervention:* none recommended on this alone
- *Priority:* **Experimental**

## 29. Drawdown Resilience Trend (*learning applied to risk, not just habit*)

- *Theory*: Learning / Exploration-Exploitation
- *Definition*: slope of Drawdown Recovery Ratio (feature 22) across an account's successive campaigns
- *Captures*: whether an account's *capacity to recover* from a bad stretch improves with experience — distinct from feature 20 (which tracks raw risk-taking drift), this tracks resilience specifically
- *Stability*: requires  $\geq 3$  qualifying campaigns
- *Business insight*: if resilience improves with tenure, that's a direct argument for retention investment (tenured traders become lower-risk to B-book against over time)
- *Intervention*: informs which tenured accounts are safe to keep B-booking vs. graduate toward A-booking treatment
- *Priority*: **Experimental**

## 30. Friction-to-Edge Ratio

- *Theory*: directly extends H2 to the individual level
  - *Definition*:  $\text{abs}(\text{total\_commission}) / \text{abs}(\text{total\_directional\_profit})$ , per account (the account-level version of the population 25 $\times$  ratio)
  - *Captures*: Stage 1 computed this at the population level and *started* the per-account version but didn't finish it ("You started this in your notebook — finish it, it's a five-line query" — the business document's own words). This is that finish: identifies which specific accounts are paying the most friction relative to their own directional skill, rather than assuming the 25 $\times$  population ratio applies uniformly.
  - *Stability*: a running ratio over an account's full trade history
  - *Business insight*: directly tests H2's own "what would validate it further" condition — whether friction drag is universal (a population law) or concentrated in a hyperactive subset (a niche)
  - *Intervention*: the fee-awareness nudge, personalized with the account's own ratio ("you've paid \$X in fees against \$Y in market-call P&L")
  - *Priority*: **Must Have**
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## 2. Rejected / Not Recommended

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- **Raw trade count, raw avg lot size, raw median duration, raw no-SL rate, raw total commission as standalone predictors of outcome** — already tested, all  $p > 0.2$ . Only their *dynamic/consistency* transformations (features 3, 9, 12, 14, 20) are worth building.
- **Round-number price anchoring, any refinement of it** — clean null result (2.53 observed vs. 2.50 expected), no reason to revisit.
- **Portfolio concentration / diversification discipline** — infeasible, single instrument.

- **Profit give-back / MAE-MFE-based features** — infeasible, no path data between open and close.
- **Shared-IP-based behavioral features** — the underlying registry field is known to be unreliable (14 rows/account, arbitrarily deduplicated); building behavioral features on top of it would compound a known data-quality problem rather than resolve it. Fix the registry pipeline first.
- **userGroupid as a cohort variable** — 99.98% one value, no comparison possible until clarified by C22.
- **Any feature requiring starting balance or entry fee** — not in the dataset; flagged for C22, not worked around.

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### 3. Ten Highest Business-Impact Features

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Ranked by direct line-of-sight to a C22 decision (early-warning system, fee-awareness nudge, or retention investment), not by statistical novelty alone.

1. **Warning-Response Simulation Score (23)** — the actual backtest for the H1 recommendation before anyone builds it.
  2. **Drawdown Recovery Ratio (22)** — tells C22 whether to intervene or let it play out.
  3. **Session Survival Time (21)** — the direct operationalization of “survival time is the product.”
  4. **Campaign Recurrence Rate (18)** — the cleanest LTV proxy in the dataset.
  5. **Friction-to-Edge Ratio, per account (30)** — finishes H2’s own unfinished validation step.
  6. **Re-entry Haste Index (6)** — the account-level test the hypothesis document explicitly calls for next.
  7. **Loss Reactivity Ratio (5)** — resolves whether “no one chases losses” survives at the individual level.
  8. **Stop-Loss Consistency volatility (9)** — resolves Stage 1’s own open question about SL usage as trait vs. mood.
  9. **Tilt Score (7)** — catches escalation Stage 1’s single-lag design structurally could not.
  10. **Overconfidence Index (25)** — the untested mirror image of the loss-chasing story, a genuinely new angle.
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### 4. Five Composite Behavioral Indices

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**A. Drawdown Resilience Index** =  $f(\text{max drawdown depth, Drawdown Recovery Ratio [22], Warning-Response Score [23], Session Survival Time [21]}) \rightarrow$  Directly extends H1 from “how deep” to “what happens next” — the core input for the early-warning system.

**B. Friction Exposure Index** =  $f(\text{Friction-to-Edge Ratio [30], Cadence Volatility [15], Session Survival Time [21], Conviction Consistency / flip-rate [11]})$  → Extends H2 to the account level — who is C22's most reliable, and least reliable, commission source.

**C. Tail Risk Index** =  $f(\text{Stop-Loss Elasticity [1], Catastrophic Exposure Rate [4], Stop-Loss Consistency volatility [9], Take-Profit Reliance Ratio [13]})$  → Extends H3 — which accounts carry the fat-tail blow-up risk the population-level test averaged away.

**D. Pressure Reactivity Index** =  $f(\text{Loss Reactivity Ratio [5], Re-entry Haste Index [6], Tilt Score [7], Post-Loss Session Abandonment [8]})$  → Extends H4 — turns “population shows haste, not size” into a per-account score.

**E. Retention Propensity Index** =  $f(\text{Campaign Recurrence Rate [18], Comeback Latency [19], Cross-Campaign Behavioral Drift [20], Drawdown Resilience Trend [29]})$  → The direct LTV/retention model input — answers which accounts are worth the CAC C22 already spent.

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## 5. Ten Testable Hypotheses Enabled by These Features

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1. Accounts with a high Re-entry Haste Index (6) show deeper eventual max drawdown than low-haste accounts — the exact Stage-2 test the hypothesis document specifies for H4, now with the feature that makes it possible.
2. The Overconfidence Index (25) predicts drawdown depth better than post-loss sizing did — testing whether pressure-driven risk-taking is really about winning streaks, not losing ones.
3. Stop-Loss Consistency volatility (9) correlates with outcome even though the raw no-SL rate does not — situational (not habitual) risk management is the real driver Stage 1's blunt 50% split couldn't detect.
4. Session Fatigue Slope on no-SL rate (14) predicts late-session catastrophic losses, independent of raw no-SL rate.
5. Campaign Recurrence Rate (18) correlates with lifetime commission paid — validating the business document's “survival time is the product” argument with a direct number.
6. Drawdown Recovery Ratio (22) predicts whether an account becomes a repeat customer — near-miss survivors are stickier than early-drawdown quitters.
7. The account-level Win/Loss Duration Asymmetry (12) shows a real disposition effect in a meaningful subset of accounts, even though the population median (295s vs. 281s) did not — resolving Stage 1's own explicitly unfinished item.
8. The Warning-Response Simulation Score (23) is materially higher for eventual losers than winners — validating that a real early-warning system would have fired with genuine lead time.
9. Exploration vs. Exploitation Score (24) correlates negatively with Friction-to-Edge Ratio (30) — habitual-size exploiters pay less avoidable commission than broad explorers.

10. Tilt Score (7) predicts top-1%-loss events better than any single-lag prior-outcome measure — proving Stage 1's lag-1 test was too coarse to catch the real escalation mechanism.

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## 6. Which Features Best Improve Prediction Of...

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**Retention / LTV:** Campaign Recurrence Rate (18), Comeback Latency (19), Cross-Campaign Behavioral Drift (20), Session Survival Time (21), Retention Propensity Index (E). These are the only features in the catalog built on the cross-campaign structure that actually encodes loyalty — everything else is single-session behavior.

**Profitability / outcome (netProfit-sign proxy):** Drawdown Recovery Ratio (22), Warning-Response Simulation Score (23), Tilt Score (7), Win/Loss Duration Asymmetry (12), Drawdown Resilience Index (A). These all extend the *one* feature Stage 1 found actually separates winners from losers (drawdown), rather than re-testing traits Stage 1 already found flat.

**Revenue / friction (commission, the business's real profit center):** Friction-to-Edge Ratio (30), Session Survival Time (21), Cadence Volatility (15), Conviction Consistency / flip-rate (11), Friction Exposure Index (B). These map directly onto the 25× friction finding — the dataset's single strongest, most robust result.

**Caveat carried from Stage 1, applying to all of the above:** every “outcome” prediction here is still against `sign(sum(netProfit))`, not the true pass/fail label — a trader could score well on every feature above and still fail the true +8%/−4% rule, or vice versa. None of these features resolve that gap; only starting-balance and breach-label data from C22 can.